

From Chiral Wave Evolution to the Principle of Logical Action

Equivalent Action, the Transcription Criterion, and Why the Bulk Quotient Is Too Coarse

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This paper supersedes “From Chiral Wave Evolution to the Principle of Equivalent Action”. That draft derived the Principle of Equivalent Action and then met three places where its equivalence relation was too coarse: the Gibbons–Hawking–York boundary term, the four-dimensional Gauss–Bonnet density, and the heat operator. It recorded each as a qualification. They are not qualifications. They are one phenomenon, and the principle that names it is the Principle of Logical Action.

The paper is in two halves separated by a printed line. Parts One to Five use standard variational calculus and elementary mathematics, and stand or fall on that. Part Six is an extension in the author’s own framework, Unified Mechanics, Corpus Edition 2026, Fifth Printing, and is offered as interpretation. Nothing above the line depends on anything below it.

A quotient is honest about what it keeps and silent about what it drops. The silence is the whole problem.

Abstract

The standard left/right decomposition of the string worldsheet wave equation is encoded on a fixed time slice as a vector in a finite mode space, evolved under a single governing operator, and used to reconstruct the complete class of actions generating that evolution. The central result of the first half is a Chiral Freeze-to-Action-Class Theorem: whenever the frozen evolution is generated by a variational operator, equivalently whenever the Helmholtz self-adjointness condition holds, the governing dynamics determine an affine equivalence class of actions unique up to null Lagrangians. The action is not the invariant dynamical object. The governing operator is, and the usual action is one representative of its class. That statement is the Principle of Equivalent Action.

The construction is then tested. It reproduces the string and wave actions, recovers the null-Lagrangian freedom by direct variation, and extends to Klein–Gordon, Maxwell and Yang–Mills with the expected gauge qualifications. Against general relativity it passes in the bulk, since Einstein–Hilbert and the first-derivative Γ – Γ Lagrangian differ by a total divergence and are two coordinates on one class. It then fails three times in a row, and the failures are the interesting part. The Gibbons–Hawking–York term distinguishes actions the bulk class identifies. The four-dimensional Gauss–Bonnet density is locally null and globally topological. The heat operator has no class at all.

The second development shows that these are not three separate qualifications. An action is a transcript of a governing operator, and a transcript is dynamically sufficient exactly when its equivalence classes are stable under the evolution one intends to run. That criterion is elementary, and it converts each failure into an instance of a single theorem: a transcript faithful to what it records

can be blind to what decides the next state. Einstein–Hilbert against Γ – Γ under a Dirichlet boundary condition is the same failure as two hotel rooms with the same brightness and different causes.

Part Six, below the line, names the principle that repairs it. The Principle of Logical Action refines equivalent-action classes by retaining order, constraints, boundary data, gauge reduction, phase, topology, measure and observational closure. Its relation to the Principle of Equivalent Action is the relation between a projection and a lift, which is the same pair the author's traversal papers describe as the circle and the helix.

Keywords: left/right movers; finite-mode freezing; governing operator; inverse calculus of variations; null Lagrangian; Helmholtz criterion; Gibbons–Hawking–York; transcription; dynamical sufficiency; Principle of Equivalent Action; Principle of Logical Action.

Attribution note. The differential equation used below is the standard two-dimensional wave equation in conformal gauge, and its left/right decomposition is d'Alembert's. It is not attributable to one physicist. Susskind was an early architect of the string interpretation of hadronic duality, and the phrase “Susskind starting point” is retained only as a conversational label for that picture.

PART I · THE FREEZE AND THE GOVERNING OPERATOR

1. The chain in one line

The whole argument is one chain:

$$\text{wave} \rightarrow \text{chiral split} \rightarrow \text{freeze on a slice} \rightarrow \text{evolve} \rightarrow \text{governing operator} \rightarrow \text{action class}.$$

The first arrow resolves the wave into two characteristic channels. The second freezes the retained state on one time slice into a fixed vector space. The third evolves that frozen state. The fourth identifies the operator governing every permitted evolution. The fifth reconstructs not one privileged action but the full class of actions having that operator as their Euler–Lagrange derivative.

The final arrow is valid exactly when the operator is variational. The Helmholtz conditions are therefore not a technical afterthought. They are the gate between evolution and action reconstruction.

2. The chiral wave starting point

Let a classical string worldsheet be parametrised by τ and σ . In conformal gauge each target-space component X obeys the two-dimensional wave equation. In characteristic coordinates $u = \tau - \sigma$ and $v = \tau + \sigma$ it reads

$$\partial_u \partial_v X = 0,$$

whose general local solution on a simply connected patch is

$$X(\tau, \sigma) = X_L(\tau + \sigma) + X_R(\tau - \sigma).$$

Proposition 2.1 (Chiral factorisation). On a simply connected patch the kernel of $\partial_u \partial_v$ consists exactly of sums $X_L + X_R$, so the two-channel description is generated by the operator $\partial_u \partial_v$.

Proof. $\partial_u \partial_v X = 0$ implies $\partial_v X$ is independent of u , so $\partial_v X = f(v)$. Integrating in v gives $X = F(v) + G(u)$. The converse is direct substitution. ■

The split is not an additional dynamical hypothesis. It is the characteristic factorisation of the governing operator. For a closed string with $\sigma \equiv \sigma + 2\pi$ the right and left towers have equal frequencies and opposite spatial momenta, and their separation is encoded by a chirality operator rather than by two unrelated equations.

3. Freezing the wave in a vector space

Freezing does not mean stopping time. It means selecting a reference slice and encoding all retained field data on that slice as one point in a fixed state space, so that evolution becomes a curve through that space.

Let Π_N be the orthogonal projector onto the first N Fourier modes, and record the retained configuration and momentum on the slice. Because Π_N commutes with the spatial wave operator, the retained subspace is invariant under the free evolution. The finite system is an exact restriction to the selected modes, not an approximate update law on them.

Collecting the complex chiral amplitudes into a vector ψ , the massless evolution is first order,

$$d\psi/d\tau = -i \Omega \Gamma \psi, \quad \psi(\tau) = \exp(-i \Omega \Gamma \tau) \psi(0),$$

with Ω storing the frequencies and Γ labelling chirality. Since $\Omega\Gamma$ is Hermitian the evolution is unitary, and since Γ commutes with it chirality is preserved: in the massless theory the two channels evolve independently.

In real mode coordinates every retained mode obeys a harmonic equation, so the frozen wave becomes an ordinary linear flow $\dot{x} = A x$. A change of coordinates $x \rightarrow Sx$ conjugates A , changing its entries and not the evolution law. The invariant content is the operator, together with the structures it preserves. In configuration space the invariant is the second-order governing operator

$$\mathbb{D}[q] = \ddot{q} + \Omega^2 q,$$

with the sign chosen so that the canonical quadratic action below has Euler–Lagrange derivative exactly $\mathcal{G}$.

This is the turning point. Once the state has been frozen into a vector space, every allowed history is generated by one operator. The action has not yet been used.

4. Reconstructing action from evolution

The chiral equation follows from a first-order action linear in $d\psi/d\tau$, and varying independently in ψ and its conjugate returns the evolution and its conjugate. That action is not unique: for any smooth $\Lambda(\psi, \bar{\psi}, \tau)$ the addition of $d\Lambda/d\tau$ leaves the bulk Euler–Lagrange equations unchanged. The equivalent-action class is therefore already visible in the frozen vector system, before any field theory is introduced.

For the second-order form the canonical quadratic action

$$S[q] = \int d\tau \quad \frac{1}{2}(\dot{q}^2 - q \cdot \Omega^2 q)$$

has Euler–Lagrange derivative $-\mathcal{G}[q]$. The frozen evolution has reconstructed a standard oscillator action without beginning from an action principle. The complex chiral action, the real phase-space action and the real configuration-space action are three presentations of one governing evolution, and their relationship is what the next part quotients.

PART II · THE PRINCIPLE OF EQUIVALENT ACTION

5. The governing operator as primitive

Let φ denote a collection of fields and let $\mathcal{D}[\varphi] = 0$ be the governing equation. The traditional direction runs from an action to equations of motion by the Euler–Lagrange map E . The reversed direction runs from governing dynamics to actions, and its output is an equivalence class rather than a single action, because E is not injective.

Definition 5.1 (Equivalent action). Two admissible local actions are equivalent in the bulk when their Euler–Lagrange expressions agree identically on the relevant jet space. This is stronger than having the same solution set: it requires equality of the governing differential operator, not merely equality of its kernel.

That the relation is an equivalence relation is immediate from equality of Euler–Lagrange maps.

6. Null Lagrangians and the affine class

A local density is null when its Euler–Lagrange expression vanishes identically. Every total divergence is null, since for compactly supported variations its contribution integrates away.

Theorem 6.1 (Local null-Lagrangian characterisation). On a contractible or star-shaped domain of finite jet space, a density is null if and only if it is a total divergence. Each bulk equivalent-action class is therefore an affine space over the vector space of total divergences.

Globally, topological densities may be closed without admitting one globally defined current. This is not a failure of the characterisation. It is the distinction between local exactness and global cohomology, and Section 13 shows it has teeth.

7. The Helmholtz gate

Not every differential equation is the Euler–Lagrange equation of an ordinary local action.

Theorem 7.1 (Helmholtz/Vainberg criterion). Let $\mathcal{D}$ be a differential operator on a star-shaped field domain. Its action class is nonempty if and only if its Fréchet derivative is formally self-adjoint, modulo integration by parts, for compactly supported test fields. When the condition holds, one representative is the Vainberg homotopy potential, which for a linear self-adjoint operator reduces to the familiar quadratic form.

After integration by parts the Vainberg representative becomes the familiar kinetic-minus-potential form plus a null Lagrangian.

8. The Chiral Freeze-to-Action-Class Theorem

Theorem 8.1. Suppose a field obeys a linear wave equation whose characteristic solutions split into left and right channels; a finite spectral projector commutes with the spatial wave operator; and the retained second-order operator is formally self-adjoint. Then the retained state on any slice is a vector, its complete evolution is generated by a single linear operator, the equivalent-action class is nonempty, a canonical representative is quadratic, and every other local representative differs from it by a null Lagrangian.

Proof. Characteristic factorisation gives the chiral split. The commuting projector preserves the retained mode subspace, so the Cauchy data define the freeze map and evolve by the linear flow. Symmetry of Ω^2 makes $\mathcal{G}$ formally self-adjoint, so the linear Helmholtz criterion applies and the Vainberg formula gives the quadratic representative. Theorem 6.1 classifies every other representative as that action plus a null Lagrangian. ■

The theorem does not say that chirality proves variationality. The split supplies a transparent operator; formal self-adjointness supplies the action. These are distinct steps, and a non-self-adjoint generator can be frozen and evolved without possessing an ordinary single-field action. Nor is freezing the assertion that the field is finite-dimensional: it is a controlled restriction, and the statement holds at every finite N and passes to the continuum whenever the truncations converge.

Principle of Equivalent Action. On the variational sector the governing operator determines a unique equivalence class of admissible actions, and the Euler–Lagrange map descends to a bijection between classes and variational operators.

PART III · THE TESTS, AND WHERE THE QUOTIENT FAILS

9. Klein–Gordon, Maxwell, Yang–Mills

For the Klein–Gordon operator formal self-adjointness is immediate and the canonical representative is the standard Lagrangian up to a divergence. The test passes exactly.

For Maxwell with a conserved source the operator is gauge-degenerate: a gradient shift of the potential changes the representative field and not the field strength. The principle therefore acts naturally after field space is quotiented by gauge transformations or after a gauge condition is imposed. This is not an extra ambiguity in the action class; it is a separate redundancy in the field coordinates, and the distinction matters for Part Five.

For Yang–Mills the nonlinear Helmholtz symmetry follows from the symmetry of the second variation, modulo gauge and boundary directions, and the standard action is one representative. The test passes on the gauge-reduced variational sector.

10. The negative control: the heat operator

Consider the heat operator $\partial_t - \kappa \partial_x^2$. Its formal adjoint is $-\partial_t - \kappa \partial_x^2$, which is not equal to it. The ordinary single-field Helmholtz criterion fails.

The frozen evolution exists perfectly well. The action class is empty. The framework therefore rejects exactly the kind of operator it should reject, and reconstruction requires added structure such as a multiplier, a response field or a doubled variable.

First failure. The transcript is empty: there is no action to be a representative of.

11. Einstein–Hilbert against Γ – Γ : the bulk pass

General relativity is the strongest test, because the bulk equations are variational, diffeomorphism gauge symmetry is present, and boundary terms carry genuine physical information.

The Ricci scalar density decomposes into a first-derivative Γ – Γ part plus a total divergence,

$$\sqrt{-g} R = \mathbb{E} \Gamma \Gamma + \partial_\mu w^\mu,$$

so the Einstein–Hilbert action, which contains second derivatives of the metric, and the Γ – Γ action, which is first order, have the same bulk Euler–Lagrange derivative. They are not competing theories. They are two coordinates on one bulk equivalent-action class, and this is the cleanest gravitational confirmation the principle has.

12. Gibbons–Hawking–York: the second failure

The decomposition of Section 11 is a bulk statement. On a manifold with boundary the variation of the Einstein–Hilbert action contains normal derivatives of the metric variation. For a Dirichlet problem in which the induced boundary metric is fixed, one adds the Gibbons–Hawking–York term.

That term does not alter the bulk Einstein tensor. It changes the boundary variational problem, and it contributes to gravitational charges and to thermodynamics. So the bulk relation is intentionally coarse, and a boundary-sensitive treatment must compare the bulk operator together with the boundary presymplectic potential and the declared boundary conditions.

The verdict is precise, and the third line of it is the one that matters:

Level	Verdict
bulk equivalent action	passes exactly
full boundary-value equivalence	requires a refined class
physical boundary data	cannot be quotiented away merely because it is invisible to the bulk Euler–Lagrange operator

Second failure. The transcript is faithful and incomplete: two actions with the same entry, and different boundary futures.

13. Four-dimensional Gauss–Bonnet: the third failure

The four-dimensional Euler density has identically vanishing bulk metric variation, subject to the usual smoothness and boundary qualifications. Adding it to any gravitational Lagrangian therefore produces the same local Einstein operator, and locally it lies in the null sector classified by Theorem 6.1.

Globally its integral measures topology and can contribute boundary terms. This is exactly the local-versus-global distinction anticipated in Section 6, and it is a second, independent witness that the bulk class discards information which is physically live.

Third failure. The transcript is locally complete and globally blind.

14. Test ledger

System	Verdict	What it establishes
Free string / wave	exact pass	the freeze-to-class chain returns the quadratic action class
Chiral first-order modes	exact pass	unitary generator, symplectic first-order class

System	Verdict	What it establishes
Klein–Gordon	exact pass	self-adjointness immediate
Maxwell	pass, gauge-reduced	degeneracy is in field coordinates, not in the class
Yang–Mills	pass, gauge-reduced	second-variation symmetry modulo gauge and boundary
Einstein–Hilbert vs Γ – Γ	exact bulk pass	one class, two coordinates
Gibbons–Hawking–York	fails	boundary data survives the quotient and matters
4D Gauss–Bonnet	fails	locally null, globally topological
Heat operator	correctly rejected	Helmholtz gate closed, class empty

Six passes and three failures. The passes were the point of the original draft. The failures are the point of this one, because they are not three.

PART IV · TRANSCRIPTION

15. Reconstruction is not assembly

The standard instinct is to separate a difficult whole into simpler parts, solve the parts, and combine the results. In sufficiently separable systems this is exact. The problem begins when the local behaviour of each part depends on relations that were omitted in order to make the parts independently calculable.

A reconstruction-first description transcribes the whole as a list of local states, advances each, and aggregates. An assembly-first description advances the complete state and obtains the local results as projections of the evolved whole. This reverses the explanatory order: reconstruction says the whole is obtained from completed parts, assembly says the parts become what they are through the evolving whole.

The claim is not that mathematics lacks fields, graphs, operators or nonlocal correlations. It contains all of them. The claim is about an architectural failure mode: a representation is permitted to discard relational distinctions during decomposition, and a later aggregation is treated as though it had restored the whole.

16. Crowley, Glorpnorp, and the transmission problem

Mr. Crowley keeps a hotel of twenty rooms. Some bulbs are broken, some rooms are dim, doors may be open or closed, walls absorb or reflect. He carries one flashlight. The easy description is room by room: record the initial illumination, the flashlight contribution, and the final illumination, then summarise.

That answer can be numerically correct and still fail to state what Crowley managed. A bright room can receive more photons and gain no useful visibility. A dark room can receive less light and become inspectable. A mirror can make a change in one room originate from a relation elsewhere. A closed door can make a geometrically pointed beam causally irrelevant.

The hotel is a weak proof of necessity, because Crowley can walk the corridors and explain what he left out. The problem becomes sharp when the receiver cannot. Glorpnorp is deaf and blind, reads only algebra through a tactile interface, and cannot inspect the hotel, infer a corridor from sight, or notice that a report looks wrong. He is a receiver with no side information.

The Crowley–Glorpnorp transmission problem. Construct an algebraic transcript that lets a receiver with no side information answer every relevant question and evolve the state correctly under the same future interventions, without seeing the system.

Human mathematical communication routinely relies on silent repair: a diagram, physical intuition, or conventional background supplies relations the equations omit. Glorpnorp forbids that repair. If two distinct conditions generate the same transcript he must treat them as the same state, and if they later respond differently the transcript was incomplete, not the reader.

17. The Field Assembly Failure Theorem

Theorem 17.1 (Field assembly failure). Let P be a decomposition or local transcription. Suppose $\mathcal{F} \neq \mathcal{G}$ with $P(\mathcal{F}) = P(\mathcal{G})$, but for some intervention I the evolutions differ, $\Phi(\mathcal{F}, I) \neq \Phi(\mathcal{G}, I)$. Then no deterministic reconstruction operating only on $P(\cdot)$ can reproduce the correct next state of both.

Proof. Because $P(\mathcal{F}) = P(\mathcal{G})$ the reconstruction receives identical input, so a deterministic map returns identical output. The correct evolutions differ, so the single output cannot equal both, and is therefore wrong for at least one. The missing distinction was not hidden inside the local values; it was removed by P , and aggregation cannot recover what its input does not contain. ■

Two corollaries follow immediately. Matching every measured local output at one time does not prove a match to the generative state, since agreement now does not imply agreement next. And a relation that participates in producing local outcomes must be present before or during the update: labelling it afterwards describes the outcome without making it causal in the model.

A sum of completed events is not the field that made the events possible.

18. The Evolution-Preserving Transcription Criterion

The failure theorem is negative. The positive requirement is stronger than ordinary reconstruction, because what is needed is not a transcript one can invert but a transcript one can evolve.

Theorem 18.1 (Evolution-preserving transcription). A transcription τ is dynamically sufficient for an evolution Φ exactly when an algebraic update Ψ exists with $\tau \circ \Phi = \Psi \circ \tau$. Writing $x \sim y$ whenever $\tau(x) = \tau(y)$, such a Ψ exists if and only if the equivalence classes induced by τ are stable under Φ .

Proof. If Ψ exists and $\tau(x) = \tau(y)$ then $\Psi\tau(x) = \Psi\tau(y)$, so by commutation $\tau\Phi(x) = \tau\Phi(y)$ and equivalent transcripts stay equivalent. Conversely, if transcript-equivalent states stay equivalent, define $\Psi(a) = \tau(\Phi(x))$ for any x with $\tau(x) = a$; stability makes the result independent of the representative, so Ψ is well defined and the square commutes. ■

This is the universal property of a quotient, stated for dynamics. Its force here is that it converts the word “meaning” into a test: a distinction is meaningful to a model exactly when removing it makes the algebraic future ambiguous. Two details may be merged when they never produce a relevant difference under the allowed interventions. They may not be merged merely because they look numerically similar now.

Theorem 18.2 (Logical state completion). If two states share the same numeric transcript but respond differently to an allowed intervention, any dynamically sufficient algebra must include at least one further distinction separating them: a predicate, relation, constraint, phase, topology or retained record.

The proof is one line: a representation containing only the numeric measurement cannot define a unique next measured state, since one input would require two outputs.

19. Three systems whose missing relations cannot be seen afterwards

The hotel is convenient. These are necessary.

Spatial. Let five regions carry identical local values in two systems, one connecting them in a line and the other connecting region one only to region five with the rest a separate chain. A local-value transcript records the same state in both. Under diffusion the next values differ, because the graph Laplacians differ. The field state is not the value vector but at least the pair of value vector and adjacency. The missing information is spatial relation itself.

Optical. Write a coherent field as amplitude times a phase factor. An intensity-only transcription records the squared amplitude. Two fields can have identical intensity at every recorded point and different phase functions, and after propagation or a lens they focus, spread or interfere differently. The intensity image was accurate and dynamically incomplete. The phase did not decorate the local amplitudes; it determined their collective future.

Quantum. The two states $(|000\rangle \pm |111\rangle)/\sqrt{2}$ have identical one-part and identical two-part reduced states, yet the three-part observable $X \otimes X \otimes X$ separates them at ± 1 . No addition of the identical local transcripts recovers the sign, because it was never a property of any part. This is the strongest form: globally distinguishing information can be absent from every permitted local transcript while remaining physically available to a joint operation.

Theorems 17.1, 18.1 and 18.2 are elementary and are stated here as ordinary mathematics. They appear in the author's Paper 17, Field Assembly and Lossless Transcription, where the hotel and the transmission problem are introduced. The spatial, optical and quantum constructions are illustrative: they establish that relational transcription loss is possible and alters prediction, not that any particular physical theory requires it.

PART V · THE TWO FAILURES ARE ONE FAILURE

20. An action is a transcript

Part Two constructed a map from actions to governing operators and quotiented by its kernel. Part Four gave the condition under which such a map may be quotiented safely. Applying the second to the first takes one line.

Let the transcription be the Euler–Lagrange map itself,

$$\tau_{PEA} : S \mapsto E[S],$$

so that two actions are transcript-equivalent exactly when they are equivalent in the sense of Definition 5.1. Theorem 18.1 then says that this transcript is dynamically sufficient for a class of operations if and only if its equivalence classes are stable under those operations.

For the operation “compute the bulk field equations”, the classes are stable by construction, and that is the content of every pass in the ledger of Section 14. For any operation that sees more than the bulk field equations, stability is a question rather than a theorem, and Sections 12 and 13 answer it in the negative.

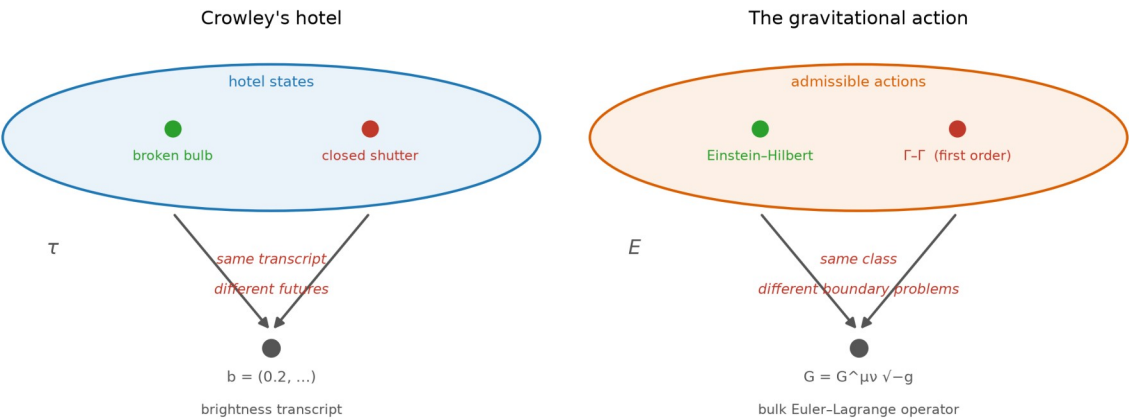
21. Gibbons–Hawking–York is a Crowley–Glorpnorp failure

Take Einstein–Hilbert and Γ – Γ . They are distinct actions. They have the same transcript under τ_{PEA} . Under a Dirichlet boundary condition their variational problems differ, their boundary presymplectic potentials differ, and their gravitational charges and thermodynamics differ. In the notation of Theorem 17.1,

$$S_{EH} \neq S_{\Gamma\Gamma}, \quad \tau_{PEA}(S_{EH}) = \tau_{PEA}(S_{\Gamma\Gamma}), \quad \Phi_{Dirichlet}(S_{EH}) \neq \Phi_{Dirichlet}(S_{\Gamma\Gamma}).$$

Those are exactly the three hypotheses of the field assembly failure theorem. The conclusion follows without further work: no procedure operating only on the bulk Euler–Lagrange operator can return the correct boundary-value problem for both.

The sentence recorded in Section 12, that physical boundary data cannot be quotiented away merely because it is invisible to the bulk Euler–Lagrange operator, is therefore not a caution. It is an instance of a theorem, and the theorem was already available.



The same failure. A transcript that is faithful to what it records, and blind to what decides the next state.

Figure 1. The same structure twice. On the left, two hotel conditions with the same brightness transcript and different responses to the same intervention. On the right, two gravitational actions with the same bulk Euler–Lagrange operator and different boundary-value problems.

The Gauss–Bonnet case is the same theorem with topology in place of boundary data, and the Maxwell case of Section 9 shows the contrast that makes the diagnosis precise. Gauge degeneracy is a redundancy in the field coordinates, which is a case where distinct descriptions genuinely have the same future and may be merged. Boundary data is not redundancy. It is a distinction that changes the future and was dropped anyway.

22. The coarseness ledger

Failure	What the transcript drops	Which theorem
Heat operator	everything: no action exists	Helmholtz gate, Theorem 7.1
Gibbons–Hawking–York	boundary data and boundary conditions	Theorem 17.1
4D Gauss–Bonnet	global topology	Theorem 17.1
Maxwell (not a failure)	gauge redundancy, correctly dropped	classes stable, Theorem 18.1

Three of the four rows are the same theorem. The first is a different one, and its difference is worth stating: the heat operator does not have an incomplete transcript, it has no transcript, because the Helmholtz gate is shut. The other two have transcripts that are faithful to what they record and blind to what decides the next state.

The Principle of Equivalent Action is a quotient. Every place it was said to require refinement is a place where the quotient is not stable under the operation being run.

That is as far as elementary mathematics carries the argument. It identifies the failures as one failure and locates it precisely. It does not say what must be retained, and it does not say why those particular things and not others. Naming that is the business of the second half.

ABOVE THIS LINE: STANDARD VARIATIONAL CALCULUS AND ELEMENTARY MATHEMATICS
BELOW THIS LINE: THE AUTHOR'S OWN FRAMEWORK, OFFERED AS INTERPRETATION

PART VI · EXTENSION · THE PRINCIPLE OF LOGICAL ACTION

23. The line, and what is below it

Everything to this point uses standard variational calculus and elementary mathematics. The Helmholtz criterion is Helmholtz's and Vainberg's. The gravitational decomposition and the boundary term are standard. Theorems 17.1, 18.1 and 18.2 are elementary. A reader who rejects everything that follows keeps all of it.

What follows is the author's own framework, and the Corpus grades are used for the remainder: PROVED follows from stated definitions or assumptions; CONDITIONAL follows once a named condition is granted, with the condition written into the statement; PROPOSED is a physical reading offered for calculation and falsification; OPEN names what can be formulated but not yet settled.

Part Five ended with two questions it could not answer. What must be retained, and why those things. The framework answers both with one principle, and the answer was written down before the failures were catalogued.

24. The Principle of Logical Action

Monograph II states the pair together:

The Principle of Equivalent Action reverses the ordinary variational direction, from action to equations of motion, into governing dynamics to an equivalence class of actions. The Principle of Logical Action then refines those action classes by retaining order, constraints, boundary data, gauge reduction, phase, topology, measure and observational closure.

Read the list against the ledger of Section 22. Boundary data is the second row. Topology is the third. Gauge reduction is the fourth, and it appears as something to be retained as a declaration rather than assumed, which is what Section 9 found when it observed that the principle acts naturally only after field space has been quotiented or a gauge fixed. The list was not assembled to fit these cases. It is the list Monograph II already carried.

The list also contains items the first half never reached, because the tests chosen did not probe them. Order and measure belong to quantisation, where the same action class can be quantised inequivalently; observational closure belongs to Part Six of Monograph II. Those are not tested here and are named so that they can be.

Restated from Monograph II, not derived here. The match between the retained list and the failure ledger of Section 22 is PROPOSED, and its strength is that three of the eight items are independently forced by Part Five without reference to the framework.

25. Equivalent action is the projection; logical action is the lift

The relation between the two principles is not that one is a longer version of the other. It is the relation between a projection and a lift, and the author's traversal papers describe the same pair under a different name.

There, a closed path is a projection of a helix, and the helix is recovered by adjoining a coordinate that records completed traversal. The projection identifies points the traversal keeps apart. Nothing about the projection is wrong; it is simply blind along the coordinate it dropped, and the blindness looks like closure.

Here, τ_{PEA} is the projection. It sends an action to its bulk Euler–Lagrange operator and identifies actions the execution keeps apart. Nothing about it is wrong. It is blind along the coordinates the execution needs, and the blindness looks like equivalence.

	Traversal papers	Here
The projection	the circle: position on the loop	τ_{PEA} : the bulk operator
What it identifies	A_0 with A_1	Einstein–Hilbert with Γ – Γ
The retained coordinate	completed traversal E	order, boundary, topology, measure
The lift	the helix	the Principle of Logical Action

	Traversal papers	Here
Why it is single	one order of accumulation	one order of execution

The last row is the one that does work. In the traversal papers the lift adjoins exactly one monotone coordinate, and the Register's T15 fixes the count by proving that the local clock mode is single, because two independent orders of accumulation would be two records and therefore two systems. The same theorem applies here. An action is executed in an order, and there is one such order, so the logical refinement is not an open-ended licence to add structure until the answer comes out right. It is the recovery of a single dropped coordinate.

Equivalent action asks what the operator is. Logical action asks what was done to obtain it. The first is a value at a point. The second is a traversal.

CONDITIONAL on identifying the order of execution of an action with the Register's single order of accumulation. Given that identification, the rank-one character of the refinement follows from T15. The identification is PROPOSED and is the load-bearing step of this part.

26. What the principle does to each failure

Boundary data. The refined comparison is not between bulk operators but between the bulk operator, the boundary presymplectic potential and the declared boundary conditions taken together. Einstein–Hilbert and Γ – Γ then separate, as they must, and the Gibbons–Hawking–York term is not a correction added to make a Dirichlet problem work. It is the part of the transcript the bulk projection dropped.

Topology. The refined class distinguishes locally null densities by their global cohomology class, so the four-dimensional Euler density is retained as a topological datum rather than discarded as null. Locally null and globally live is exactly the situation Section 6 anticipated and Section 13 exhibited.

Gauge. The framework's treatment here is a declaration rather than a discovery, and it should be graded that way. Gauge reduction appears in the retained list, meaning that which reduction was performed is part of the transcript. This is consistent with Section 9, where the classes turned out to be stable precisely once the reduction was declared, and it is the one item in the list where the first half found stability rather than failure.

The heat operator is not repaired by this principle, because it is not a refinement problem. It is treated separately in the next section, where it closes by the other posture.

The boundary and topology readings are PROPOSED and are supported by the independent derivation of Part Five. The gauge reading is a declaration.

27. The heat operator, closed

The heat operator was listed in Section 22 as a different kind of failure and left there. It can be closed, and closing it uses the lift rather than the refinement.

Ask where the Helmholtz obstruction actually lives. For $\mathcal{D} = \partial_t - \kappa \partial_x^2$ the Fréchet derivative is $\mathcal{D}$ itself, and its formal adjoint is $-\partial_t - \kappa \partial_x^2$. The spatial term is untouched, being even in ∂_x and therefore self-adjoint. The entire discrepancy sits in the one term that is odd under reversal of t .

That is general rather than accidental. Under integration by parts $(\partial_t)^\dagger = -\partial_t$, so a term of even order in the order coordinate is unchanged by taking the adjoint and a term of odd order changes sign. Therefore:

The Helmholtz obstruction lives entirely in the part of the operator that is odd under reversal of the order coordinate, and nowhere else.

Three cases settle the reading. The wave operator is second order in t , even under reversal, and passes with a single real field. The heat operator is first order in t with a single real field and fails. The Schrödinger operator is also first order in t and passes, and the reason it passes is the point: its field is complex, so the system is already a pair, and the standard Lagrangian $i\bar{\psi}\partial_t\psi - (1/2m)|\partial_x\psi|^2$ works because varying $\bar{\psi}$ gives the equation while varying ψ gives its conjugate, which runs in the opposite order direction. Schrödinger is not an exception to the criterion. It already carries both directions of the order.

This identifies what the heat operator lacks and what the standard repair supplies. The Martin–Siggia–Rose construction adjoins a response field and writes

$$S = \int dt dx \quad \tilde{\varphi} (\partial_t \varphi - \kappa \partial_x^2 \varphi).$$

Varying $\tilde{\varphi}$ returns the heat equation. Varying φ returns $-\partial_t \tilde{\varphi} - \kappa \partial_x^2 \tilde{\varphi} = 0$, the same equation run in the opposite order direction. The doubled system is self-adjoint as a pair, and it is so precisely because the second field carries the direction the first cannot.

That is the traversal lift, in a different domain and with the same shape. The base space cannot hold the order, so the base object has no action; adjoin one coordinate that carries the order and the action exists. It is also the same count: one field adjoined, not two, because there is one order to carry.

So the heat operator is not a fourth kind of failure. The other two failures were transcripts that dropped something they should have kept. This is an object that cannot be transcribed at all until it is lifted, because it has content along exactly the coordinate the action formalism quotients away. The Helmholtz gate is not an arbitrary admissibility test. It is the question of whether the order survives the projection.

The location of the Helmholtz obstruction in the order-odd sector is PROVED, by inspection of the adjoint, and the three cases are standard. The identification of the Martin–Siggia–Rose doubling with the traversal lift is PROPOSED, supported by the two constructions having the same form and the same

rank rather than by a derivation of either from the other. This moves the heat operator from OPEN to PROPOSED.

28. What this answers, and what it owes

	Left open by the first half	What carries it	Grade
1	Which failures are the same failure	Theorems 17.1 and 18.1	PROVED
2	What must be retained	PLA retained list	PROPOSED
3	Why those items and not others	single order of execution	CONDITIONAL
4	Why the refinement is not open-ended	T15, rank one	CONDITIONAL
5	The heat operator	obstruction is order-odd; lift supplies it	PROPOSED

One identification is owed, and it is named in Section 25: that the order in which an action is executed is the Register's single order of accumulation. Sections 25, 26 and 27 rest on it and nothing else does. If it fails, the retained list survives as a list and loses its claim to be forced rather than assembled, which is a real loss and not a fatal one.

What would end this part. A dynamically relevant distinction dropped by the bulk quotient that is not on the retained list would show the list incomplete. A member of the retained list that never separates two bulk-equivalent actions under any admissible operation would show it padded. The second is the easier test and has not been run.

29. The shared identification

This paper and the traversal papers are separate arguments in separate domains, and it is worth recording that their open items are not separate.

The traversal paper owes the identification of its index A_0, A_1, A_2 with the Register's single order of accumulation. This paper owes the identification of the order of execution of an action with the same thing. Those are one identification stated twice, in two vocabularies. Granting it once grants it for both; refusing it once refuses it for both.

Paper	What it calls the order	Status
The Depth of Traversal	traversal index	one identification
This paper	order of execution	the same one
The Rate Across the Scale	closure per e-fold	a different one, a rate

The third is genuinely separate, being a proportionality between two counts rather than an identity between two orders, and it carries its own observational handle: if closures were not in fixed proportion to e-folds the invariant η_H would run, and its running is measured to be -0.0023 ± 0.0034 per e-fold. That is a weak bound and it is a bound, which is more than the other identification has.

So across three papers the ledger is two open items, not four. One is an identity and is untested. The other is a proportionality and is bounded. Stating that plainly is worth more than three separate confessions that look independent and are not.

30. Closing

The earlier draft derived one principle and met three obstacles. It recorded them carefully, which is why they could be recognised, and it treated them as separate qualifications on a result rather than as evidence about the result's shape.

They were one thing. A quotient had been taken, and the quotient was stable for the operation it was designed for and unstable for every other. Elementary mathematics is enough to see that, and it is not enough to say what the quotient should have kept.

The Principle of Logical Action says what it should have kept, and says it as a list that was written before these particular failures were catalogued. Three of its items are independently forced by the first half of this paper. That is the strongest thing that can be claimed for it here, and it is claimed at the grade the Corpus uses for readings of this kind.

Equivalent action recovers the operator and forgets the execution. Logical action is the execution kept. The first is what the projection reports; the second is what was done.

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